Shanice Reih H. Tanque

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Education

Mindanao State University - Iligan Institute of Technology — Iligan City, Lanao del Norte
Bachelor of Science in Computer Applications — 2023 - 2027
Major in Embedded Systems
GPA: 1.32

Mindanao State University at Naawan - Integrated Developmental School — Naawan, Misamis Oriental
Science, Technology, Engineering, and Mathematics — 2017 - 2023
With Honors

Experience

Lead Generation Assistant — April 2025 - December 2025

- Researched, verified and organized prospective client and company contact data to support lead-generation outreach and follow-up.

Research Extension Division, MSU at Naawan — Naawan, Misamis Oriental
Work Immersion Intern — January 2023 - April 2023

- Supported Panguil Bay water-quality testing, microscope-based species identification and laboratory research data collection.

Projects

YOLOv8n Bell Pepper Aphid-Damage Detection on Raspberry Pi 5 (In Progress)

- Developing an offline Raspberry Pi 5 and YOLOv8n system to identify aphid-related damage in bell pepper leaves with local image-based monitoring.

RT-Spark Cloud-Based Human Activity Recognition

- Co-developed a walking-versus-running recognition system using RT-Spark motion data, MQTT and a Random Forest cloud pipeline, contributing embedded firmware and Wi-Fi bring-up.

RT-Spark Personal Music Player

- Built and tested an STM32F407ZG FreeRTOS music player with I2S/DMA audio, an ES8388 codec, potentiometer volume control and LCD feedback.

FreeRTOS Multisensor Room Monitoring System

- Implemented an STM32F103C8T6 room monitor coordinating five FreeRTOS tasks for environmental sensing, OLED display, rotary input, motion detection and alarms.

Industrial Drying Oven PLC-HMI Control System

- Simulated a CODESYS Structured Text oven controller with a five-state process, HMI setpoints, temperature regulation, door interlocks and emergency shutdown.

VERTANOM - Smart Vertical Farming System

- Co-developed a Qt/C++ dashboard with SQLite for monitoring soil moisture, pH, temperature and humidity in a smart vertical farming prototype.

Arduino ATmega328P Line-Following Robot

- Developed and documented an Arduino ATmega328P-based line-following robot to follow a marked route through autonomous control.

AVR Microcontroller Assembly Programming Labs

- Implemented ATmega328P assembly exercises involving registers, subroutines, stack operations, binary search, coin arithmetic, seven-segment displays and digital I/O.

BrainBoard - RT-Thread Interactive Quiz

- Developed an RT-Thread and LVGL quiz interface featuring event-driven controls, multiple screens, timed questions and score tracking.

AquaFlush Water-Flow Monitoring

- Built an Arduino Uno water-flow monitoring prototype using interrupt-based pulse counting and an I^2C LCD to display measurements.

Touch-Sensitive Coin Tosser

- Designed and tested a touch-triggered transistor circuit and PCB prototype that produces red or green LED coin-toss outcomes.

BCA131 Analog Electronics Laboratory Series

- Completed eight circuit laboratories exploring BJT biasing, transistor switching, LED drivers and common-emitter amplifiers with documented testing.

Technical Skills

Programming Languages: C, C++, AVR Assembly, Structured Text (IEC 61131-3), SQL
Embedded Systems & IoT: STM32, FreeRTOS, RT-Thread, Raspberry Pi 5, Arduino/ATmega328P, ESP32, Sensor Integration, Edge IoT
Industrial Automation: CODESYS, PLC Programming, HMI Development, Finite State Machines, Safety Interlocks
Frameworks & Tools: STM32Cube HAL, Qt, LVGL, PlatformIO, Visual Studio Code, Arduino IDE, Wokwi, Git, GitHub
Hardware & Interfaces: I2S, DMA, ADC, PWM, I2C, OLED/LCD, ES8388 Audio Codec, Sensors and Actuators
Databases & Testing: SQLite, MySQL, Unity Test Framework, Cppcheck, Host and Functional Testing

Leadership & Activities

College of Computer Studies - Student Council, MSU-IIT — Iligan City, Lanao del Norte
Undersecretary, Ways and Means Committee — August 2025 - May 2026

- Supported student council resource planning and organizational coordination as Ways and Means Undersecretary, earning leadership and service commendations.

College of Computer Studies - Student Council, MSU-IIT — Iligan City, Lanao del Norte
Co-Chairperson, Culture and Arts Committee — August 2024 - May 2025

- Co-led Culture and Arts activities and college events, coordinating student council initiatives and receiving recognition for committee service.

myComApps Technology Symposium & Exhibit - RT-Thread-Based Technology — Iligan City, Lanao del Norte
Overall Committee Coordinator & Attendance Committee Member — November 2025

- Coordinated myComApps symposium committees and attendance while exhibiting BrainBoard, an RT-Thread and LVGL embedded quiz application.

my.ComApps Technology Development Program — Iligan City, Lanao del Norte
Head, Attendance and Certificate Committee — December 2024

- Led participant attendance tracking and certificate preparation and distribution for the my.ComApps Technology Development Program.

Achievements

EUSPN 2026 - Accepted Full Paper (Co-author) — Almaty, Kazakhstan
17th International Conference on Emerging Ubiquitous Systems and Pervasive Networks — October 2026

- Third co-author of accepted EUSPN 2026 full paper #75, “Computational Approaches, Evidence Maturity, and Research Gaps in Bell-Pepper Disease and Pest Monitoring: A Systematic Literature Review,” alongside Tristan Listanco, Phoebien Cañete, Apple Rose Alce and Paul Rodolf Castor.

Sustainable Innovation Pitch 2024 - Champion — Iligan City, Lanao del Norte
MSU-IIT Sustainable Development Day — September 2024

- Won Champion at MSU-IIT Sustainable Innovation Pitch 2024 for the VERTANOM smart vertical farming system.

TechShowcase 2025 - IoT System for Aphids Damage Detection in Bell Pepper Farms — Iligan City, Lanao del Norte
MSU-IIT — December 2025

- Earned Best in Pitching (Champion) and Best in Innovative System (1st Runner-Up) for the bell pepper pest-damage monitoring project.

Certificate of Commendation - CCC102 Computer Programming 2 — Iligan City, Lanao del Norte
College of Computer Studies, MSU-IIT — March 2025

- Recognized for creative programming projects, technical presentations, mentorship and contributions supporting fellow students.

Certifications & Training

Internet of Things Conference 2025 - Guest — September 2025

- Attended the two-day Internet of Things Conference 2025 at SMX Convention Center, Pasay City, as a guest.

Startup Basics 101 - iDEYA:CIT, MSU-IIT — September 2026

- Completed an iDEYA:CIT session covering startup fundamentals, innovation and entrepreneurship.

AI Career Readiness Training - ASEAN Ahead, ASEAN Foundation — August 2026

- Completed ASEAN Foundation / ASEAN Ahead training in career readiness and practical AI skills.